

Fall Detection Using 1D CNN

This project uses a 1D Convolutional Neural Network (1D CNN) to classify human postures and movement patterns — specifically **Sitting**, **Standing**, and **Walking** — using time-series data from a 24GHz radar sensor. The system aims to contribute to robust and lightweight fall detection systems in smart health and elderly care applications.

Current Accuracy

- **98%** test accuracy achieved on 3-class classification (Sitting, Standing, Walking)
 - Dataset created using real-time radar readings
 - Model trained on windowed sequences from `Presence_Byte_4` and `Energy_Byte_8` features
-

Dataset

Each CSV file includes:

- `timestamp`
- `Presence_Byte_4`
- `Energy_Byte_8`
- `Expected outcome`
- `label` (manually assigned or inferred)

Files:

```
- sitting_training.csv
- standing_training.csv
- walking_training.csv
- sitting_testing.csv
- standing_testing.csv
- walking_testing.csv
```

Model Architecture

A simple 1D CNN architecture with:

- 2 Convolutional layers
- MaxPooling
- Dropout
- Dense + Softmax output

Framework: **TensorFlow / Keras**

How to Run

1. Clone the repository:

```
git clone https://github.com/yourusername/fall-detection-1dcnn.git
cd fall-detection-1dcnn
```

1. Install dependencies:

```
pip install -r requirements.txt
```

1. Run the training script:

```
python train.py
```

Confusion Matrix

After training, a confusion matrix is displayed to show classification performance.

Future Work

- Add **fall** as a fourth class
- Integrate real-time sensor input with a microcontroller
- Export model to TensorFlow Lite for edge deployment

Project Structure

```
♦ data/
├─ sitting_training.csv
├─ standing_training.csv
├─ ...
♦ train.py
♦ README.md
♦ requirements.txt
```

Contributing

PRs and feedback are welcome! If you'd like to improve the fall detection pipeline, feel free to fork and contribute.



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